Enterprise Software Origination and Underwriting Case

An illustrative growth capital underwriting of a hypothetical B2B enterprise SaaS company, comparing growth equity, private credit, and blended capital against the same operating case.

Important disclosure

Northstar Workflow Systems is a hypothetical company created solely for an illustrative underwriting exercise. It is not a real business or investment opportunity.

1. Executive Summary

Northstar Workflow Systems begins with USD 12.0m of ARR, 30% Year 1 growth, 110% net revenue retention, 88% gross retention, and 78% gross margin. The company requires USD 20.0m of capital to refinance USD 3.0m of existing debt, fund the operating plan, and maintain liquidity through the growth period.

Preliminary base-case recommendation

Use USD 8m of primary equity and USD 12m of senior secured recurring-revenue debt. The blend reduces dilution to 7.7%, leaves USD 8.1m of Year 5 cash, and produces a 2.54x equity MOIC. It is conditional: the downside case ends with only USD 0.4m and breaches the USD 5.0m minimum cash threshold.

Metric	Growth equity	Private credit	Selected blend
Equity invested	USD 20.0m	USD 0.0m	USD 8.0m
Debt principal	USD 0.0m	USD 20.0m	USD 12.0m
Founder dilution	17.2%	0.0%	7.7%
Year 5 cash	USD 16.0m	USD 2.3m	USD 8.1m
Minimum cash breach	None	Year 4	None
Equity investor MOIC	2.44x	n/a	2.54x
Equity investor IRR	19.5%	n/a	20.5%
Debt investor IRR	n/a	12.1%	11.4%
Downside Year 5 cash	USD 8.4m	USD (5.4)m	USD 0.4m

Company and transaction assumptions

Operating assumption	Value	Transaction assumption	Value
Beginning ARR	USD 12.0m	Beginning cash	USD 8.0m
Year 1 ARR growth	30.0%	Existing debt	USD 3.0m
Net revenue retention	110.0%	Entry ARR multiple	8.0x
Gross retention	88.0%	Exit ARR multiple	7.0x
Gross margin	78.0%	Minimum cash	USD 5.0m

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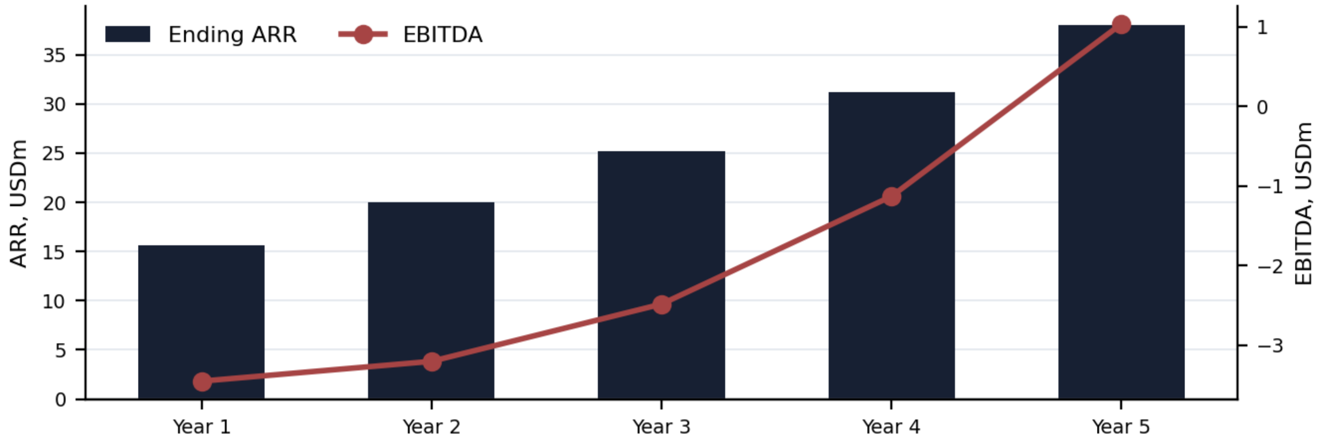
2. Operating Case and SaaS Quality

The forecast is built from an ARR bridge rather than a top-down revenue growth assumption. Churn and expansion are set by gross and net retention. New ARR is the residual required to reach the growth target, making the sales burden explicit.



USD millions	Year 1	Year 2	Year 3	Year 4	Year 5
Beginning ARR	12.0	15.6	20.0	25.2	31.2
New ARR	2.4	2.8	3.2	3.5	3.7
Expansion ARR	2.6	3.4	4.4	5.5	6.9
Churned ARR	(1.4)	(1.9)	(2.4)	(3.0)	(3.7)
Ending ARR	15.6	20.0	25.2	31.2	38.1
Revenue	13.8	17.8	22.6	28.2	34.6
EBITDA	(3.5)	(3.2)	(2.5)	(1.1)	1.0
Unlevered FCF	(3.4)	(3.1)	(2.4)	(1.1)	1.0

SaaS quality metrics

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
ARR growth	30.0%	28.0%	26.0%	24.0%	22.0%
Net revenue retention	110.0%	110.0%	110.0%	110.0%	110.0%
Gross retention	88.0%	88.0%	88.0%	88.0%	88.0%
Gross margin	78.0%	78.0%	78.0%	78.0%	78.0%
CAC payback, months	23.2	22.8	22.0	21.1	20.1
Burn multiple	0.93x	0.71x	0.47x	0.18x	n/a
Rule of 40	5	10	15	20	25
Customers	55	68	84	105	131
ACV, USD k	n/a	n/a	n/a	n/a	n/a

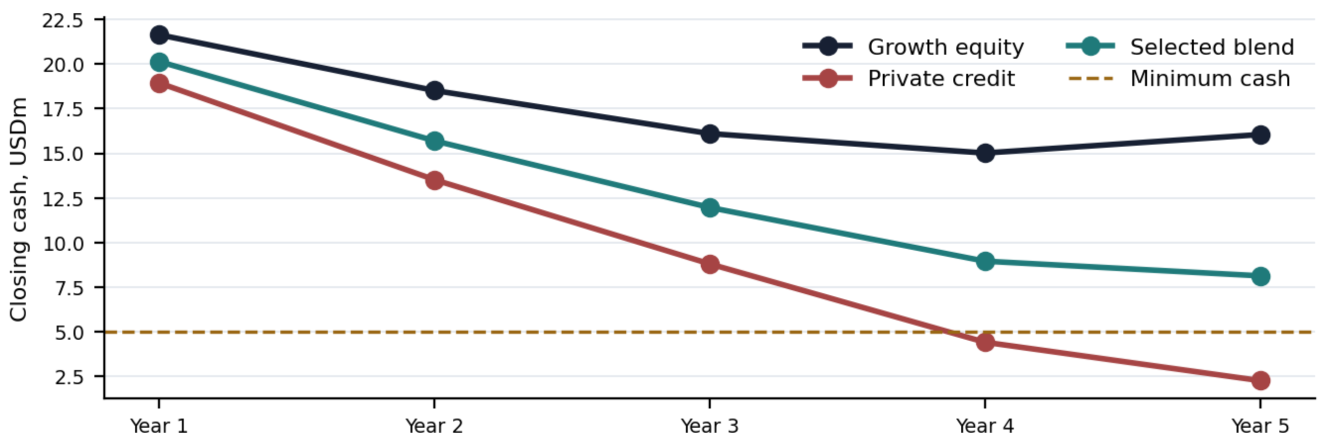
What supports the case: NRR of 110% exceeds gross retention of 88%, so the installed base expands without new logos. The burn multiple improves from 0.93x to 0.18x, and EBITDA reaches positive territory in Year 5.

What constrains it: Rule of 40 begins at 5 and reaches only 25. Gross retention of 88% means 12% of the base is lost each year, and CAC payback remains around two years.

3. Capital Need and Structure Economics

The USD 20 million capital plan is larger than the cumulative operating burn alone. The sizing bridge includes debt refinancing, modeled burn, debt service, minimum cash, existing liquidity, and original issue discount. Product, sales, international expansion, and working capital are already embedded in the operating forecast, so they are not counted again as separate uses.

Capital sizing bridge - selected blend	USD millions
Existing debt refinanced	3.0
Base case cumulative operating burn	9.0
Base case debt service	7.7
Minimum ending cash	5.0
Less beginning cash	(8.0)
OID / financing friction	0.2
Base case gross capital required	16.9
Gross capital raised	20.0
Base case excess headroom	3.1
Downside gross capital required	24.6
Downside funding shortfall	4.6



Credit framing

Recurring-revenue facility, not a conventional cash-flow loan

EBITDA is negative through Year 4, so debt service coverage remains below 1.0x. The credit case must therefore be sized primarily to recurring revenue, minimum liquidity, customer concentration, and retention. DSCR is shown as a warning metric and refinancing risk remains material at maturity.

Credit metric	Year 1	Year 2	Year 3	Year 4	Year 5
Debt to ending ARR	0.77x	0.60x	0.48x	0.37x	0.28x
Interest coverage	-2.61x	-2.43x	-1.88x	-0.85x	0.83x
Debt service coverage	-2.82x	-2.69x	-2.22x	-0.88x	0.19x
Closing cash	20.1	15.7	12.0	9.0	8.1
Covenant headroom	15.1	10.7	7.0	4.0	3.1

4. Structure Alternatives and Holder Value

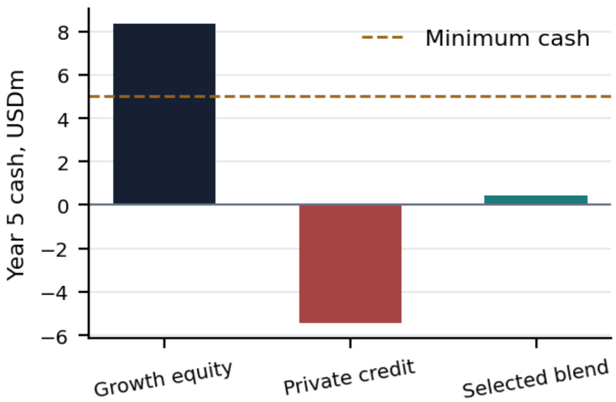
Case	Core economics	Key advantage	Primary weakness
Growth equity	USD 20.0m equity; 17.2% dilution	No debt service or covenant risk	Existing holders give up 17.2% ownership
Private credit	USD 20.0m debt at 11.5%	No dilution	Year 4 liquidity breach; USD 18.0m remains at maturity
Selected blend	USD 8.0m equity + USD 12.0m debt	Dilution falls to 7.7%	Downside liquidity breach; USD 10.8m remains at maturity

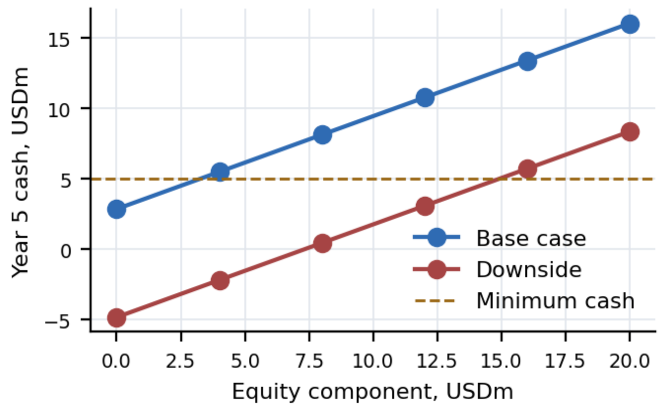
Exact existing holder value

Metric	Growth equity	Private credit	Selected blend
Existing holder ownership	82.8%	100.0%	92.3%
Exit equity value	USD 282.5m	USD 250.7m	USD 263.8m
Existing holder proceeds	USD 233.8m	USD 250.7m	USD 243.5m
Incremental versus all equity	USD 0.0m	USD 16.9m	USD 9.7m

Effect on existing holders

Relative to the all-equity structure, the selected blend increases modeled proceeds to existing holders by approximately USD 9.7 million while reducing dilution from 17.2% to 7.7%.





Equity / debt mix sensitivity

Equity	Debt	Dilution	Base Y5 cash	Downside Y5 cash	Status
USD 0m	USD 20m	0.0%	USD 2.9m	USD (4.8)m	FAIL
USD 4m	USD 16m	4.0%	USD 5.5m	USD (2.2)m	FAIL
USD 8m	USD 12m	7.7%	USD 8.1m	USD 0.4m	BREACH
USD 12m	USD 8m	11.1%	USD 10.8m	USD 3.1m	BREACH
USD 16m	USD 4m	14.3%	USD 13.4m	USD 5.7m	Headroom
USD 20m	USD 0m	17.2%	USD 16.0m	USD 8.4m	Headroom

The selected 8 / 12 mix is the base-case return-maximizing blend. At whole-dollar increments, at least USD 15m of equity is required to preserve the downside minimum cash threshold. The first tested four-million-dollar step with downside headroom is USD 16m equity and USD 4m debt.

5. Downside Case and Key Risks

The downside case begins with 18% growth, 100% NRR, 82% gross retention, 74% gross margin, and an additional 5% of sales and marketing expense. It is designed to test whether the capital structure remains viable when operating leverage is delayed.

USD millions	Year 1	Year 2	Year 3	Year 4	Year 5
Ending ARR	14.2	16.4	18.7	21.0	23.1
Revenue	13.1	15.3	17.6	19.8	22.0
EBITDA	(4.4)	(4.1)	(3.5)	(2.6)	(1.3)
Unlevered FCF	(4.5)	(4.2)	(3.6)	(2.8)	(1.6)
Growth equity cash	20.5	16.3	12.7	9.9	8.4
Private credit cash	17.8	11.3	5.4	(0.7)	(5.4)
Selected blend cash	19.0	13.5	8.5	3.8	0.4

Boundary of the recommendation

The all-equity case ends the downside with USD 8.4m and no liquidity breach. The all-debt case fails at USD (5.4)m. The selected blend survives with USD 0.4m but breaches minimum cash in Year 4. A decision-maker who places meaningful probability on this downside should increase the equity component or require a committed liquidity backstop.

Key risks

Risk	Assessment
Operating leverage	Operating expenses fall from 103% of revenue in Year 1 to 75% in Year 5. A slower decline keeps EBITDA negative and makes debt more restrictive.
Gross retention	GRR of 88% implies 12% annual base loss. The downside uses 82%, forcing more new-logo ARR when cash is tight.
Rule of 40	The company reaches only 25 by Year 5. The 7.0x exit multiple should be treated as an upper-end assumption for this profile.
Refinancing risk	USD 10.8m remains outstanding at maturity against only USD 1.0m of Year 5 unlevered FCF.
Covenant headroom	Base-case headroom declines to USD 3.1m. Downside headroom turns negative in Year 4.
Category bundling	Large application vendors may bundle workflow capability into existing suites, limiting pricing power and raising retention risk.

Preliminary conclusion

The selected blend is appropriate only as a preliminary base-case structure. It funds the plan, reduces dilution from 17.2% to 7.7%, and increases existing-holder proceeds by USD 9.7m versus all equity. It does not remove downside risk. Final sizing should be revisited after management provides cohort retention, contracted ARR, customer concentration, monthly cash burn, and the full debt schedule.

6. Additional Diligence and Model Controls

Area	What must be confirmed
Retention	Three years of cohort-level GRR and NRR, split by contract size, product, and geography.
Revenue quality	Contracted ARR versus usage or month-to-month revenue, contract length, renewal timing, and multi-year mix.
Customer concentration	Top 10 and top 20 customer revenue, sector concentration, and renewal calendar.
Unit economics	Fully loaded CAC by segment and channel, expansion cost, implementation burden, and payback by cohort.
Cost structure	Bottom-up headcount and non-headcount plan supporting the decline in operating expense ratios.
Existing debt	Principal, interest, amortization, security, covenants, prepayment penalties, and any hidden liabilities.
Working capital	Billing frequency, deferred revenue, days sales outstanding, collections, and annual prepayment behavior.
Gross margin	Subscription versus services mix, cloud infrastructure costs, implementation margin, and support burden.
Capital plan	Exact timing of product, sales, international, and working-capital uses; acquisition spend is not separately modeled.

Model control summary

Check	Status
Base ARR bridge	OK
Revenue convention	OK
Capital structures total	OK
Growth equity cash roll	OK
Private credit cash roll	OK
Blended cash roll	OK
Debt principal continuity	OK
Exit equity bridge	OK
Existing holder value	OK
Downside selected cash	OK
All-debt minimum cash breach	OK
Selected blend base headroom	OK
Selected blend downside warning	OK
Summary outputs are formulas	OK

Disclosure

Northstar Workflow Systems is hypothetical. All operating, financial, and transaction assumptions are illustrative and do not represent an actual company or investment recommendation. This is an independent work sample built by Sahil Modi and is not affiliated with or endorsed by any investment firm.

Sahil Modi designed the research framework, scoring logic, underwriting structure, and investment analysis. AI-assisted development tools were used to support coding, research organization, testing, and document production. The accompanying Excel model contains editable assumptions, live formulas, scenario sensitivities, and formula-driven integrity checks.

Accompanying workbook: Enterprise Software Growth Capital Model.xlsx
Public repository: github.com/smodi13/growth-capital-origination-engine
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